Ather EV Sales & Operations Dashboard

End-to-End Business Intelligence & Data Analytics Case Study

Executive Summary

This project delivers a complete end-to-end analytics solution designed to simulate real-world decision-making for an electric vehicle business. The primary objective is to analyze sales performance, customer behavior, product trends, and operational efficiency through a structured data model and an interactive dashboard.

The dataset has been synthetically generated using Python, based on research-driven business assumptions such as seasonal demand variation, evolving product mix, financing behavior, and the impact of delivery timelines on customer satisfaction.

The final solution is designed for deployment and portfolio presentation, enabling stakeholders to interactively explore insights and make data-driven business decisions.

Business Performance Analysis | Maharashtra | 2024–2025

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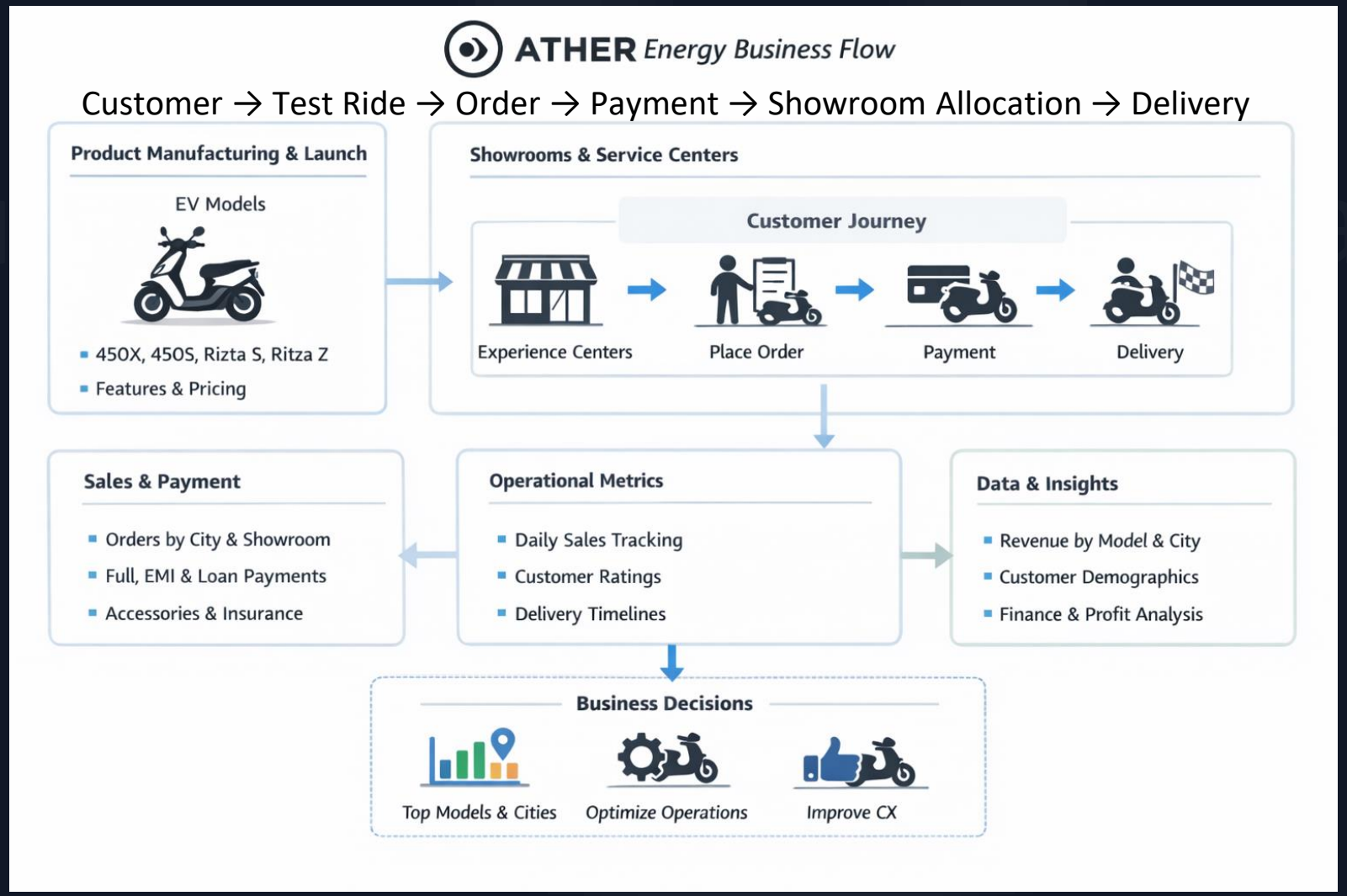
End-to-End Business Process Flow

Overview of the complete business lifecycle from customer interaction to final delivery.
This slide represents the full business journey starting from customer interest through test ride, order placement, payment processing, and final delivery.
It highlights how different entities such as customers, products, showrooms, and operations interact within the system.
Understanding this flow is critical to interpreting sales performance, financial metrics, and operational efficiency.



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Business Problem Statement

The electric vehicle business lacks a centralized analytics system to monitor and optimize performance.
Key challenges include:
•Limited visibility into sales performance across regions
•Lack of understanding of customer purchasing behavior
•Inefficient tracking of product performance
•Operational gaps in delivery timelines
•Inability to measure impact of customer satisfaction on business outcomes

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Key Business Insights

- Sales show strong seasonal patterns, with peak demand during festive months and decline during monsoon periods
- Loan and EMI-based purchases dominate, indicating high customer price sensitivity
- Newly launched models are gaining higher adoption, reflecting shifting market preferences
- Faster delivery timelines are directly associated with higher customer satisfaction ratings
- Tier-2 cities are showing consistent growth, indicating strong expansion potential
- Discounts and accessory bundling significantly influence final purchase value

Business Recommendations

- Align inventory and marketing strategies with seasonal demand spikes
- Optimize delivery operations to maintain turnaround time within 5 days
- Strengthen financing partnerships to improve customer affordability
- Focus promotional efforts on newly launched models
- Expand showroom presence in high-growth Tier-2 cities
- Implement targeted pricing and bundling strategies to maximize revenue

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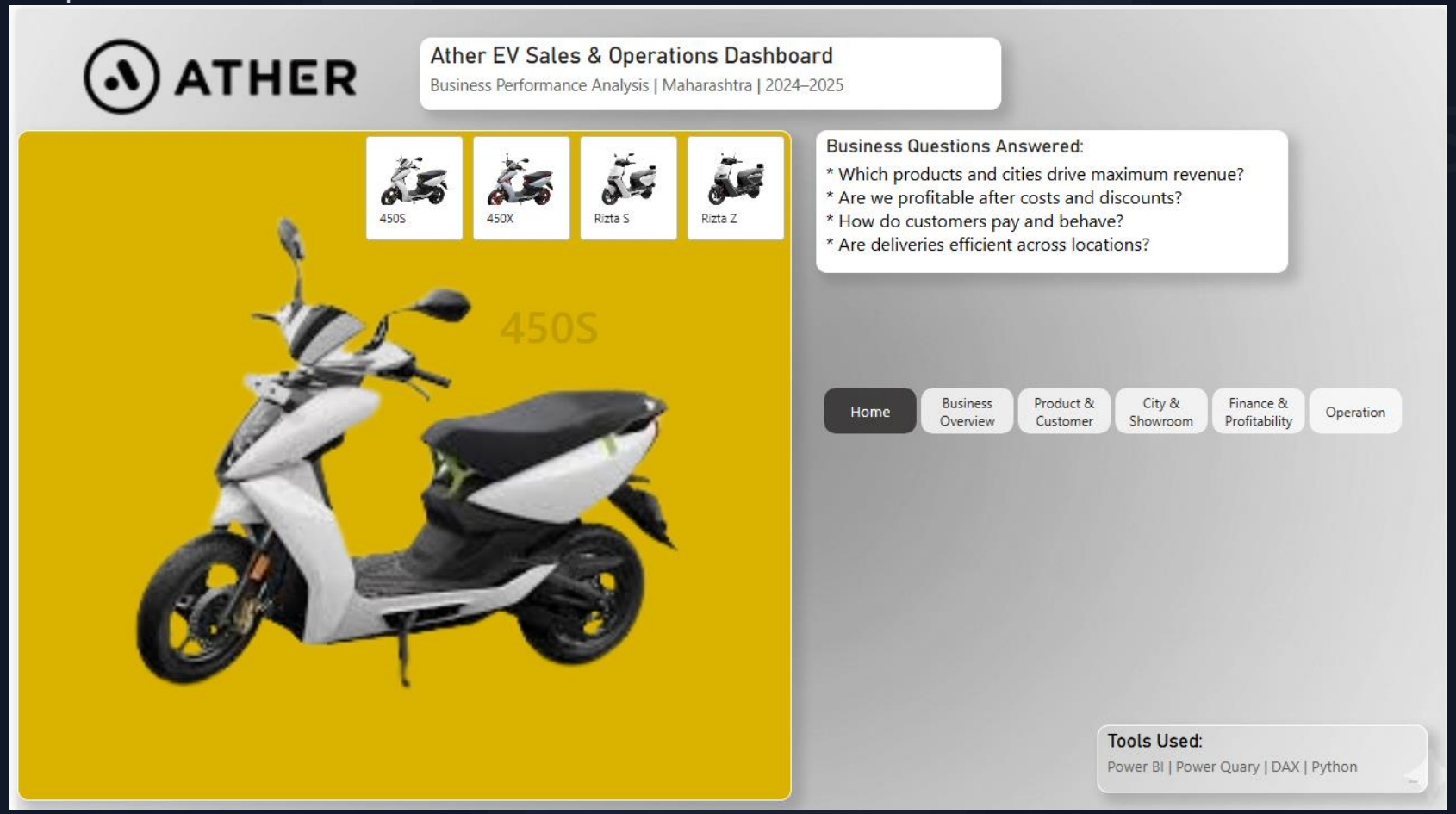
Dashboard Overview

The Power BI dashboard is designed as a multi-page analytical solution providing a comprehensive view of business performance.
Each dashboard page is supported with screenshots to ensure clear visual understanding and effective communication with stakeholders.
The dashboard enables:

- Real-time KPI monitoring
- Trend analysis across time and geography
- Deep-dive into product and customer behavior



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Business Performance Overview

High-level view of revenue, profit, demand trends, and overall business growth across time and regions.

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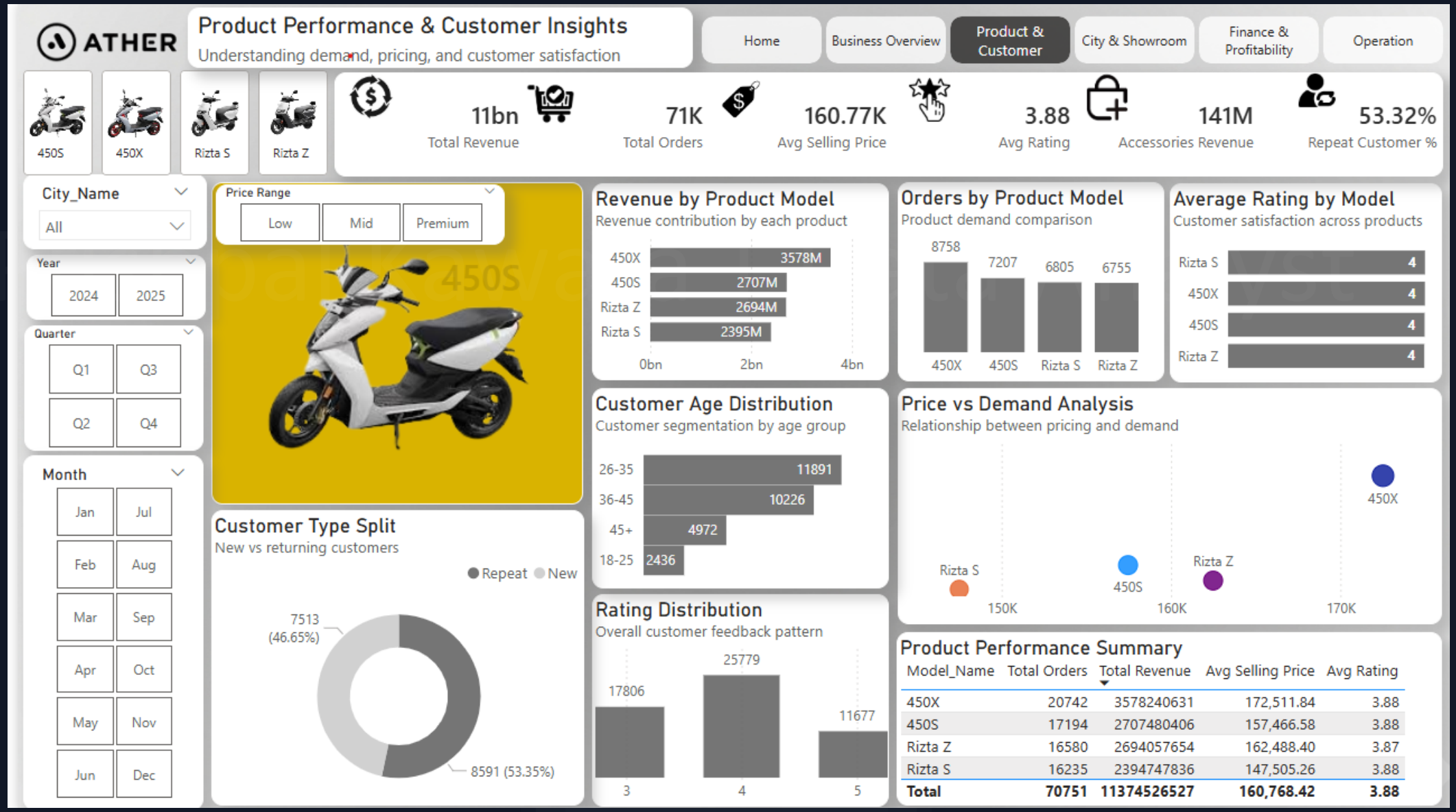
Product Performance & Customer Insights

Analysis of product demand, pricing impact, and customer behavior to understand buying patterns.



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Geographic & Showroom Performance

Evaluation of city-wise demand and showroom efficiency to identify top-performing locations.

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Geographic & Showroom Performance

City-level demand and showroom efficiency analysis

Home | Business Overview | Product & Customer | City & Showroom | Finance & Profitability | Operation

450S | 450X | Rizta S | Rizta Z

11bn	71K	Pune	Thane 44	1.18K	189.58M
Total Revenue	Total Orders	Top City	Top Showroom	Avg Orders Per Showroom	Revenue per Showroom

City_Name: All

Showroom_N...: All

Year: 2024 | 2025

Quarter: Q1 | Q2

Month: Jan | Jul | Feb | Aug | Mar | Sep | Apr | Oct | May | Nov | Jun | Dec

Price Range: Low | Mid | Premium

Top Cities by Revenue

City-wise revenue comparison

Pune	969M
Solapur	968M
Nagpur	966M
Mumbai	962M
Thane	948M
Jalgaon	947M
Chhatra...	945M
Nashik	941M
Kolhapur	940M
Nanded	940M

Orders by Showroom

Showroom performance comparison

Nande...	540
Sangli 20	531
Sangli 31	528
Jalgao...	524
Nashik ...	522
Sangli 1	518
Mumb...	517
Thane 44	517
Mumb...	514
Amrav...	512
Chhatr...	512

Revenue Efficiency by Showroom

Revenue generated per showroom

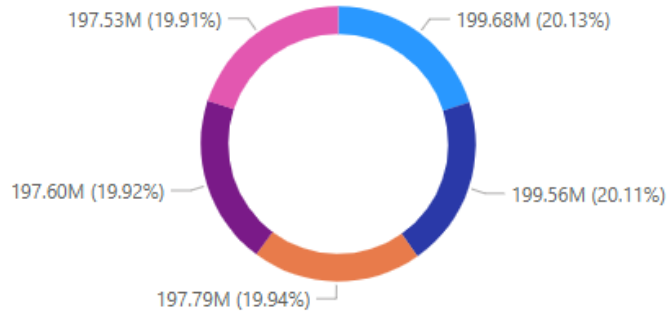
Nande...	86.68M
Sangli 20	85.16M
Sangli 31	84.59M
Jalgao...	84.11M
Nashik ...	83.91M
Mumb...	83.24M
Sangli 1	83.18M
Thane 44	82.76M
Mumb...	82.65M
Solapu...	82.63M

City & Showroom Performance

City_Name	Showroom_Name	Total Orders	Total Revenue	Revenue per Showroom
Amravati	Amravati 13	92	14880080	14,880,080.00
Amravati	Amravati 23	91	14680864	14,680,864.00
Amravati	Amravati 24	95	15220223	15,220,223.00
Amravati	Amravati 40	104	16596873	16,596,873.00
Amravati	Amravati 5	92	14739795	14,739,795.00
Amravati	Amravati 57	82	13182964	13,182,964.00
Amravati	Chhatrapati Sambhajinagar 21	107	17032533	17,032,533.00
Amravati	Chhatrapati Sambhajinagar 25	93	14837278	14,837,278.00
Amravati	Chhatrapati Sambhajinagar 36	91	14759350	14,759,350.00
Total		**70751**	**11374526527**	**189,575,442.12**

Showroom Contribution

Share of total revenue by showroom



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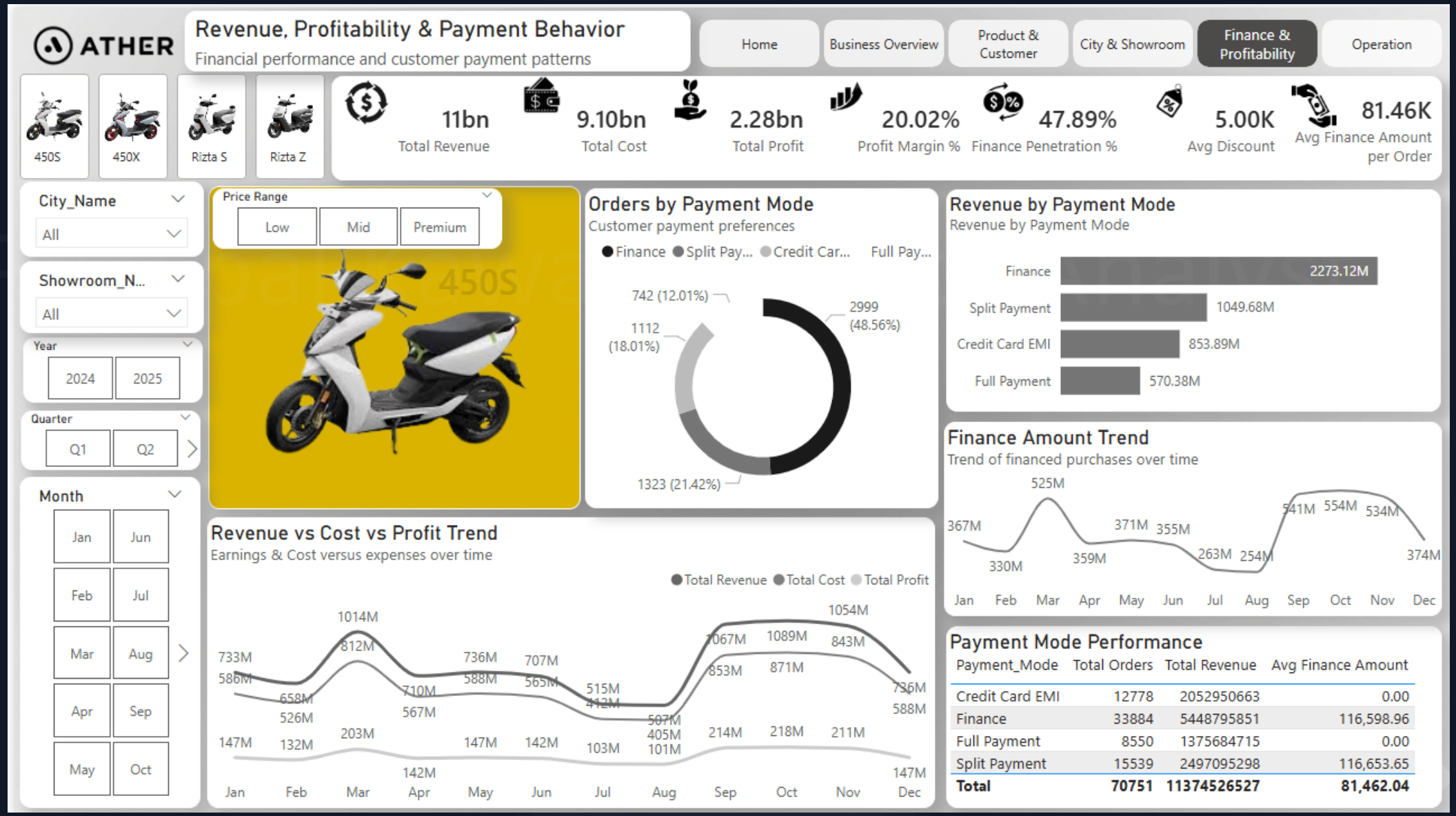
Financial Performance & Payment Behavior

Breakdown of revenue, cost, profit, and customer payment preferences to assess financial health.



Payment_Mode	Total Orders	Total Revenue	Avg Finance Amount
Credit Card EMI	12778	2052950663	0.00
Finance	33884	5448795851	116,598.96
Full Payment	8550	1375684715	0.00
Split Payment	15539	2497095298	116,653.65
Total	**70751**	**11374526527**	**81,462.04**

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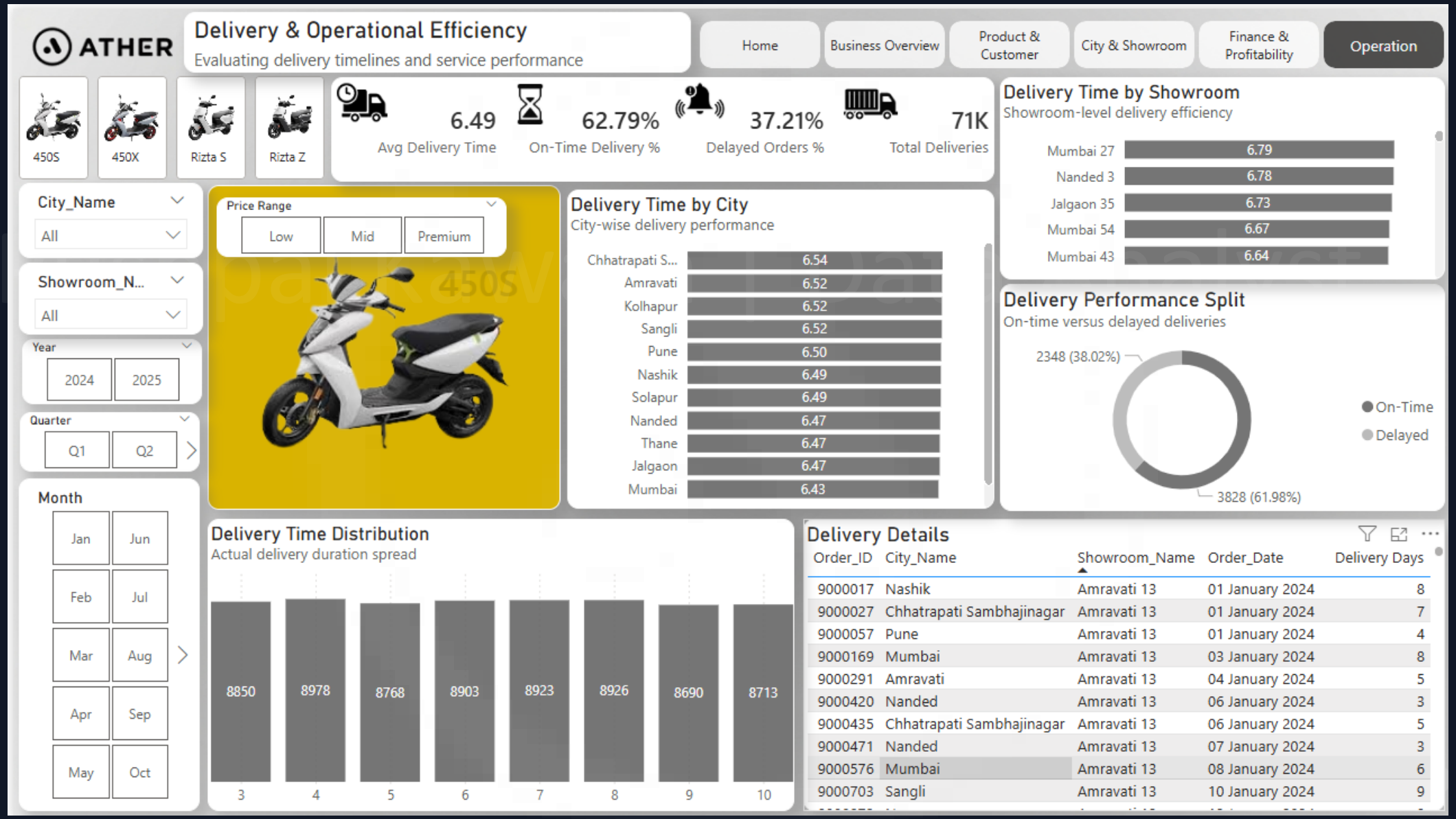
Delivery & Operational Efficiency

Assessment of delivery timelines, service performance, and operational effectiveness.



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Data Understanding

The dataset is generated using a Python-based simulation model designed to replicate real-world business conditions.

Data Coverage:

- Time Period: January 2024 – December 2025
- Customers: ~52,000
- Cities: Multiple locations across Maharashtra
- Showrooms: 60+

Data Structure:

Fact Table:

- Fact_Sales (transaction-level data)

Dimension Tables:

- Dim_Customer
- Dim_Product
- Dim_City
- Dim_Showroom
- Dim_Payment
- **Dim_Date (created using DAX for time intelligence)**

Data Generation Logic

The dataset is built using research-based assumptions to ensure realistic behavior.

- Seasonal demand variation (festive spike, monsoon dip)
- Product mix evolution between 2024 and 2025
- High dependency on financing options
- Delivery time influencing customer satisfaction
- Pricing derived from multiple real-world components

This approach ensures meaningful and business-relevant analysis.

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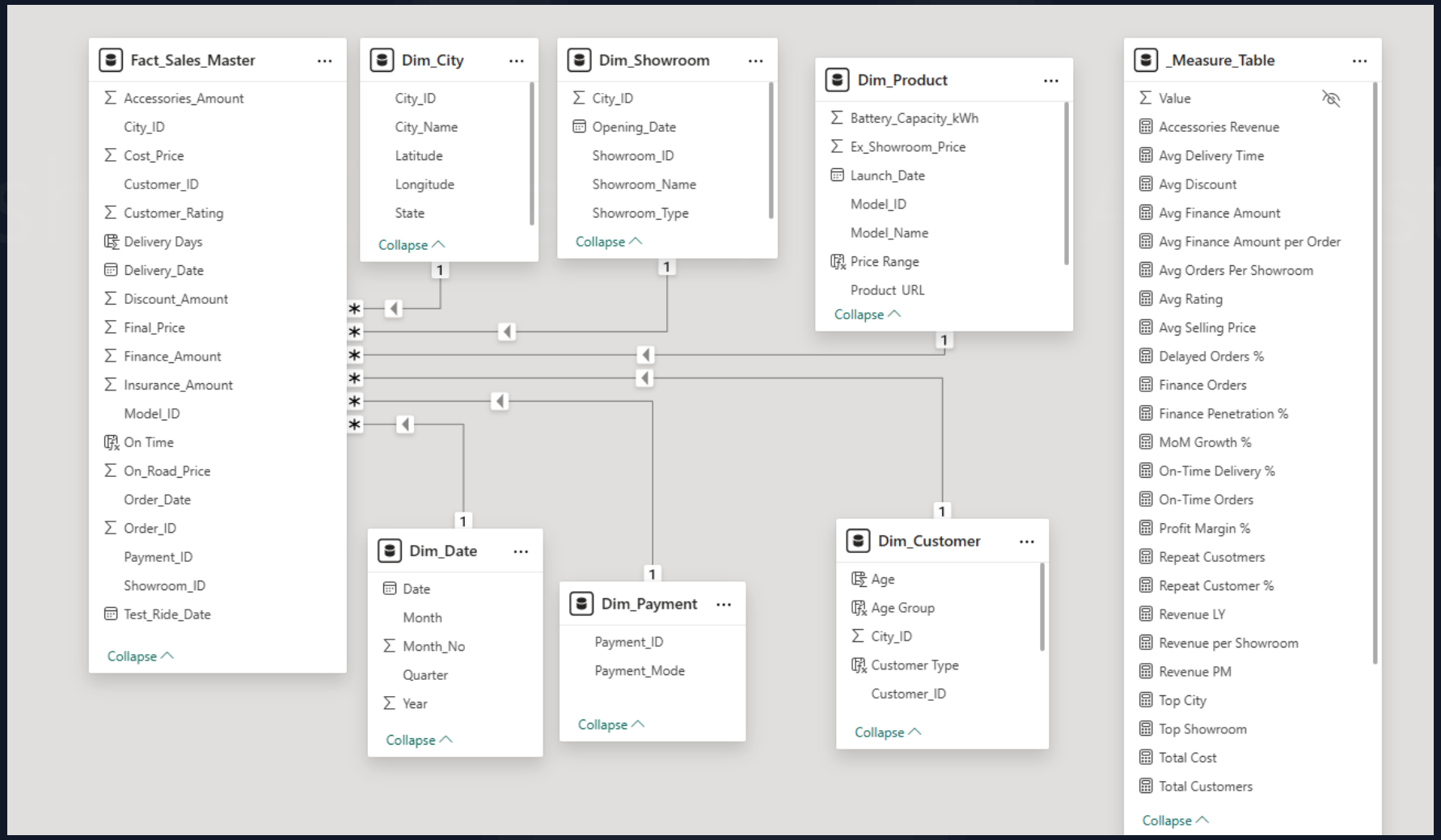
Data Model (Star Schema)

A star schema model is implemented for efficient analytics.

- Central Fact Table: Sales transactions
- Linked Dimension Tables: Customer, Product, City, Showroom, Payment, Date

Benefits:

- Faster performance
- Clear relationships
- Scalable design



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Data Preparation (Power Query + DAX Date Table)

Data cleaning and transformation were performed using Power Query.

Key steps:

- Handling missing values
- Data standardization
- Creating derived columns
- Maintaining referential integrity

A **Dim_Date table was created using DAX** to support time-based calculations such as YoY and MoM analysis.

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KPI & Measures (DAX Layer)

Key metrics developed using DAX:

- Total Sales Revenue
- Total Orders
- Average Order Value
- Total Profit
- Profit Margin %
- Average Delivery Time
- Customer Rating
- Finance Contribution %

These KPIs provide a comprehensive performance view across all business dimensions.

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Tools, Technologies & Deployment

- Python (Data Generation & Simulation)
- Excel (Data Exploration)
- Power BI (Dashboard Development)
- Power Query (Data Transformation)
- DAX (Calculations & Measures)
- PowerPoint (Presentation & Communication)

Deployment:

- Dashboard prepared for deployment and embedded into a portfolio website
- Interactive access enables stakeholders and clients to explore insights dynamically

Limitations

- Dataset is synthetically generated and based on assumptions
- External real-world factors such as competition and policy changes are not included
- Results may differ in actual business scenarios

Conclusion

This project demonstrates a complete end-to-end analytics workflow, covering data generation, modeling, transformation, and visualization.

The analysis is built on research-driven assumptions and realistic business logic, ensuring that the insights generated are meaningful, practical, and aligned with real-world decision-making scenarios.

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